AIGP Module 2: AI Impacts on People and Responsible AI Principles

Comprehensive Study Notes

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1. INTRODUCTION TO AI IMPACTS

1.1 Understanding AI Harms

AI systems, while offering significant benefits, can cause various types of harm to individuals, groups, organizations, and society. Understanding these harms is essential for implementing effective governance.

Key Principle: AI harms are not always intentional—they often result from:

- Biased training data
- Flawed design choices
- Inadequate testing
- Misalignment between technical goals and societal values
- Unforeseen interactions with real-world contexts

Governance Imperative: Organizations must proactively identify, assess, and mitigate potential harms throughout the AI lifecycle.

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2. CATEGORIES OF AI HARMS

2.1 Individual Harms

Definition: Negative impacts affecting specific persons through AI system decisions or operations.

Types of Individual Harms:

Discrimination and Bias:

- Unfair treatment based on protected characteristics
- Denial of opportunities (employment, credit, housing)
- Differential accuracy across demographic groups
- Stereotyping and profiling

Privacy Violations:

- Unauthorized data collection or use
- Re-identification of anonymized data

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- Inference of sensitive attributes
- Surveillance and tracking

Economic Harm:

- Unfair pricing or terms
- Wrongful denial of benefits or services
- Financial losses from errors
- Reduced employment opportunities

Physical and Psychological Harm:

- Safety risks from autonomous systems
- Emotional distress from unfair treatment
- Manipulation of behavior
- Loss of autonomy

Examples:

- **Amazon's biased hiring tool**: Discriminated against women because trained on historical data showing male preference in tech roles
- **Healthcare AI inaccuracies**: Algorithms performing worse for certain demographics, leading to misdiagnosis
- **Credit scoring bias**: Unfair denial of loans to minority applicants
- **Facial recognition errors**: Higher error rates for people with darker skin tones, leading to false accusations

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2.2 Group Harms

Definition: Negative impacts affecting entire demographic groups, communities, or populations.

Types of Group Harms:

Systemic Discrimination:

- Perpetuation of historical inequalities
- Reinforcement of stereotypes
- Exclusion of marginalized communities
- Amplification of existing biases

Representational Harm:

- Misrepresentation or erasure of groups

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- Reinforcement of harmful stereotypes
- Lack of cultural sensitivity
- Tokenization or oversimplification

Distributional Inequity:

- Unequal distribution of AI benefits
- Concentration of harms in vulnerable populations
- Digital divide and access disparities
- Resource allocation biases

Examples:

- **Predictive policing**: Disproportionate targeting of minority neighborhoods based on biased historical data
- **Insurance algorithms**: Systematic disadvantaging of certain racial or socioeconomic groups
- **Educational AI**: Perpetuating achievement gaps through biased assessment tools
- **Language models**: Underperformance on dialects and languages of marginalized communities

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2.3 Societal Harms

Definition: Broad negative impacts affecting society as a whole, social institutions, or democratic processes.

Types of Societal Harms:

Erosion of Trust:

- Loss of confidence in institutions
- Reduced trust in information ecosystems
- Skepticism toward technology
- Social fragmentation

Misinformation and Manipulation:

- Generation and spread of false information
- Deepfakes undermining truth
- Political manipulation
- Propaganda and influence campaigns

Democratic Threats:

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- Election interference
- Manipulation of public opinion
- Erosion of informed consent
- Concentration of power

Social Division:

- Filter bubbles and echo chambers
- Polarization and radicalization
- Erosion of shared reality
- Weakening of social cohesion

Labor Market Disruption:

- Job displacement and unemployment
- Skill obsolescence
- Wage pressure
- Economic inequality

Examples:

- **Deepfakes**: Videos manipulating political figures' images and voices to spread disinformation
- **Social media algorithms**: Amplifying divisive content and creating echo chambers
- **Automated content moderation**: Suppressing legitimate speech or failing to catch harmful content
- **AI-generated disinformation**: Large-scale creation of fake news and propaganda

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2.4 Organizational Harms

Definition: Negative impacts on organizations deploying or affected by AI systems.

Types of Organizational Harms:

Reputational Damage:

- Public backlash from AI failures
- Loss of customer trust
- Brand damage
- Negative media coverage

Legal and Regulatory Risk:

- Non-compliance penalties

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- Litigation costs
- Regulatory sanctions
- Criminal liability

Operational Failures:

- System errors and downtime
- Poor performance and unreliability
- Integration challenges
- Dependency on flawed systems

Financial Losses:

- Costs of remediation
- Lost business opportunities
- Reduced market value
- Insurance claims

Examples:

- **Microsoft's Tay chatbot**: Had to be shut down within 24 hours due to inappropriate outputs
- **Tesla Autopilot incidents**: Reputational and legal challenges from accidents
- **Wells Fargo and others**: Regulatory scrutiny over algorithmic discrimination
- **Clearview AI**: Legal challenges over privacy violations in facial recognition

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2.5 Environmental Harms

Definition: Negative impacts on the natural environment and climate from AI development and deployment.

Types of Environmental Harms:

Energy Consumption:

- Massive electricity use for training large models
- Ongoing inference costs
- Data center energy demands
- Carbon footprint

Resource Depletion:

- Mining for specialized hardware components
- Water usage for cooling data centers

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- Electronic waste from hardware obsolescence
- Rare earth mineral extraction

Climate Impact:

- Greenhouse gas emissions
- Contribution to global warming
- Environmental degradation
- Sustainability challenges

Scale:

- Training a single large language model can emit as much carbon as several cars over their lifetimes
- Data centers account for significant global energy consumption
- Growing AI usage increases environmental pressure

Governance Responses:

- Energy-efficient algorithms and hardware
- Renewable energy for data centers
- Model efficiency optimization
- Carbon accounting and offset programs
- Responsible hardware lifecycle management

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3. TYPES AND SOURCES OF BIAS IN AI

3.1 Understanding Bias in AI

Definition: Systematic and unfair discrimination against certain individuals or groups in favor of others.

Why Bias Matters:

- Perpetuates and amplifies societal inequalities
- Undermines fairness and justice
- Erodes trust in AI systems
- Can violate laws and regulations
- Causes tangible harm to individuals and groups

3.2 Types of Bias

Implicit Bias (Human Bias):

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Definition: Unconscious attitudes, stereotypes, or preferences held by humans that influence AI development.

How It Enters AI:

- Developers' unconscious assumptions in design choices
- Labelers' subjective judgments in creating training data
- Product managers' biases in defining success metrics
- Researchers' choices in problem formulation

Examples:

- Defining "creditworthy" based on traditional banking norms that disadvantage certain groups
- Labeling images with culturally specific interpretations
- Choosing features that correlate with protected characteristics
- Prioritizing use cases that benefit majority populations

Mitigation:

- Diverse, inclusive teams
- Awareness training on unconscious bias
- Structured decision-making processes
- External audits and reviews

Sampling Bias (Data Bias):

Definition: Training data that doesn't accurately represent the population the AI will serve.

How It Occurs:

- Historical underrepresentation of certain groups
- Convenience sampling from readily available data
- Geographic or demographic limitations in data collection
- Selection bias in data sources

Examples:

- **Facial recognition trained primarily on lighter-skinned faces**: Poor performance on darker skin tones
- **Medical AI trained on data from specific demographics**: Reduced accuracy for other populations
- **Language models trained predominantly on English text**: Poor performance in other languages

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- **Hiring AI trained on historical successful hires**: Perpetuates existing workforce composition

Impact:

- Higher error rates for underrepresented groups
- Reduced effectiveness for minority populations
- Unfair outcomes and discrimination
- Exclusion from AI benefits

Mitigation:

- Representative data collection
- Oversampling underrepresented groups
- Synthetic data generation
- Regular bias audits
- Diverse data sources

Temporal Bias (Concept Drift):

Definition: Model performance degradation when the world changes after training, causing the model's learned patterns to become outdated or invalid.

How It Occurs:

- Social norms and behaviors evolve
- Economic conditions change
- New patterns emerge
- Population demographics shift
- Language and terminology evolve

Examples:

- **Credit scoring models**: Economic recession changes what predicts default
- **Fraud detection**: Fraudsters adapt tactics, old patterns become obsolete
- **Hiring AI**: Job market and skill requirements evolve
- **Medical diagnosis**: New treatments or disease prevalence changes
- **COVID-19 impact**: Sudden shifts in consumer behavior, work patterns, etc.

Impact:

- Increasing error rates over time
- Outdated decisions based on obsolete patterns

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- Loss of relevance and effectiveness
- Unfair treatment based on outdated assumptions

Mitigation:

- Continuous monitoring for drift
- Regular model retraining
- Adaptive learning systems
- Version control and updating schedules
- Trigger-based retraining when performance drops

Overfitting:

Definition: Model learns the training data too well, including noise and idiosyncrasies, failing to generalize to new data.

How It Occurs:

- Model is too complex for the amount of training data
- Training for too many iterations
- Insufficient diversity in training data
- Memorizing specific examples rather than learning general patterns

Symptoms:

- Excellent performance on training data
- Poor performance on test/real-world data
- Inability to handle novel situations
- Brittle behavior with slight variations

Examples:

- Medical diagnosis system that memorizes specific patient records but fails on new patients
- Image classifier that works on training images but fails on similar real-world images
- Chatbot that repeats training examples verbatim but can't handle variations

Impact:

- Unreliable predictions in real-world deployment
- False confidence in model performance
- Potential for serious errors
- Wasted resources on ineffective systems

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Mitigation:

- Regularization techniques
- Cross-validation during training
- Larger, more diverse training datasets
- Simpler model architectures
- Early stopping in training
- Ensemble methods

3.3 Measurement and Detection of Bias

Approaches to Bias Detection:

Statistical Analysis:

- Compare error rates across demographic groups
- Analyze outcome distributions
- Test for disparate impact
- Measure fairness metrics

Fairness Metrics:

- **Demographic parity**: Equal positive outcome rates across groups
- **Equalized odds**: Equal true positive and false positive rates
- **Predictive parity**: Equal positive predictive value across groups
- **Individual fairness**: Similar individuals receive similar predictions

Auditing Methods:

- Pre-deployment bias testing
- Red teaming and adversarial testing
- Third-party audits
- Ongoing monitoring post-deployment

Challenges:

- No single definition of fairness
- Tradeoffs between different fairness metrics
- Limited access to demographic data
- Intersectionality (multiple protected characteristics)
- Context-specific determinations

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4. PRIVACY RISKS AND PROTECTION

4.1 Privacy Risks in AI

Data Collection Risks:

- Excessive data gathering beyond necessity
- Lack of informed consent
- Secondary use of data for unintended purposes
- Collection of sensitive attributes

Data Processing Risks:

- Inference of sensitive information not directly collected
- Re-identification of anonymized data
- Data linkage across sources
- Profiling and categorization

Data Sharing Risks:

- Third-party access to personal data
- Cross-border transfers
- Data breaches and unauthorized access
- Vendor risks

Algorithmic Risks:

- Surveillance and tracking
- Behavioral prediction and manipulation
- Privacy violations through AI outputs
- Model inversion attacks (reconstructing training data)

4.2 Privacy-Enhancing Technologies (PETs)

Definition: Technologies that protect privacy while enabling data use and AI functionality.

Differential Privacy:

What It Is: Mathematical technique adding carefully calibrated noise to data or query results to protect individual privacy while preserving aggregate patterns.

How It Works:

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- Adds statistical noise to data or outputs
- Guarantees that removing any single individual's data doesn't significantly change results
- Provides mathematically provable privacy bounds

Key Concept: Privacy budget (epsilon) - measures privacy loss

- Lower epsilon = stronger privacy protection but less accuracy
- Higher epsilon = weaker privacy but better utility

Applications:

- Census data release
- Medical research databases
- Query responses on sensitive datasets
- Model training on private data

Advantages:

- Mathematically rigorous privacy guarantee
- Resistant to linkage attacks
- Enables data sharing with privacy protection

Limitations:

- Privacy-utility tradeoff
- Cumulative privacy loss over multiple queries
- Difficulty choosing appropriate privacy parameters
- May reduce accuracy for small subgroups

Example: U.S. Census Bureau uses differential privacy to protect individual responses while publishing aggregate statistics.

Homomorphic Encryption:

What It Is: Encryption method allowing computations on encrypted data without decrypting it.

How It Works:

- Data remains encrypted throughout processing
- Computations performed on ciphertext produce encrypted results
- Results can be decrypted only by authorized parties
- Original data never exposed to processing party

Types:

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- **Partially homomorphic**: Supports limited operations (e.g., only addition or only multiplication)
- **Somewhat homomorphic**: Limited number of operations
- **Fully homomorphic**: Arbitrary computations (most powerful but computationally expensive)

Applications:

- Cloud computing on sensitive data
- Medical data analysis
- Financial calculations
- Privacy-preserving machine learning

Advantages:

- Strong security even if processing server is compromised
- Enables outsourced computation without data exposure
- Compliance with strict privacy regulations

Limitations:

- Computationally intensive (slower than plaintext operations)
- Complex to implement
- Limited practicality for some real-time applications
- Requires significant computational resources

De-identification (Anonymization and Pseudonymization):

Anonymization:

What It Is: Irreversible removal or modification of identifying information so individuals cannot be re-identified.

Techniques:

- Removing direct identifiers (names, addresses, IDs)
- Aggregation and suppression
- Generalization (e.g., exact age → age range)
- Data perturbation

Standard: Must meet high threshold—truly irreversible

Legal Status:

- Under GDPR, properly anonymized data is no longer "personal data"

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- Falls outside scope of data protection regulations
- No consent or legal basis required

Challenges:

- Difficult to achieve true anonymization
- Risk of re-identification through data linkage
- Advances in technology increase re-identification risks
- Balancing utility and privacy

Pseudonymization:

What It Is: Replacing identifying information with pseudonyms or tokens; reversible with additional information held separately.

How It Works:

- Replace direct identifiers with pseudonyms/codes
- Maintain separate lookup table (kept secure)
- Can re-identify if needed for legitimate purposes

Legal Status:

- Under GDPR, pseudonymized data remains "personal data"
- Subject to data protection regulations
- Recognized as a security measure but not complete anonymization

Use Cases:

- Research requiring re-identification capability
- Healthcare records linking
- Data sharing with privacy safeguards

Advantages:

- Reduces privacy risk while maintaining data utility
- Enables legitimate re-identification when needed
- Easier to implement than full anonymization

Federated Learning:

What It Is: Machine learning approach where models are trained across decentralized devices/servers holding local data, without exchanging the data itself.

How It Works:

1. Central server distributes initial model to participants

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2. Each participant trains model on their local data
3. Only model updates (gradients/parameters) are sent to central server
4. Server aggregates updates to improve global model
5. Updated global model redistributed to participants
6. Process repeats iteratively

Key Principle: "Bring the model to the data, not the data to the model"

Applications:

- Mobile keyboard prediction (Google's Gboard)
- Healthcare AI across hospitals without sharing patient data
- Financial fraud detection across banks
- IoT device learning

Advantages:

- Data remains on local devices/servers
- Reduces privacy risks from centralized data storage
- Enables collaboration without data sharing
- Complies with data localization requirements
- Reduces bandwidth for data transfer

Limitations:

- Communication costs for model updates
- Heterogeneity of data across participants
- Potential for model inversion or membership inference attacks
- Coordination complexity
- Slower convergence than centralized training

Privacy Enhancements:

- Combined with differential privacy for stronger guarantees
- Secure aggregation protocols
- Encryption of model updates

4.3 Privacy Principles in AI Governance

Data Minimization:

- Collect only data necessary for specified purpose

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- Avoid "just in case" data collection
- Regular review and deletion of unnecessary data

Purpose Limitation:

- Use data only for stated, legitimate purposes
- Obtain new consent for new purposes
- Clear documentation of intended uses

Storage Limitation:

- Retain data only as long as necessary
- Automated deletion policies
- Regular data retention reviews

Transparency:

- Clear privacy notices
- Disclosure of AI use
- Information about data processing

User Rights:

- Access to personal data
- Correction of inaccuracies
- Deletion/erasure rights
- Opt-out of certain processing

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5. SECURITY RISKS IN AI

5.1 Types of AI-Specific Security Threats

Model Inversion Attacks:

What It Is: Attacker reconstructs training data or sensitive information by querying the model and analyzing outputs.

How It Works:

- Repeatedly query model with crafted inputs
- Analyze prediction patterns
- Reconstruct information about training data
- Extract sensitive attributes

Risk: Exposure of private information from training data

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Example: Reconstructing faces from facial recognition model, inferring medical conditions from health prediction model

Mitigation:

- Differential privacy during training
- Query rate limiting
- Output perturbation
- Access controls

Model Extraction (Stealing):

What It Is: Attacker creates a functionally equivalent copy of a proprietary model by observing its inputs and outputs.

How It Works:

- Query target model extensively
- Use responses to train surrogate model
- Replicate functionality without access to original training data or architecture

Risks:

- Intellectual property theft
- Circumventing access controls and fees
- Enabling further attacks on stolen model
- Competitive disadvantage

Mitigation:

- Query monitoring and rate limiting
- Watermarking model outputs
- Authentication and access controls
- Legal protections (contracts, IP law)

Data Poisoning:

What It Is: Attacker manipulates training data to compromise model behavior.

Types:

Availability Attacks:

- Goal: Reduce overall model performance
- Method: Inject mislabeled or corrupted data

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Integrity Attacks (Backdoor Attacks):

- Goal: Cause specific misbehavior on trigger inputs
- Method: Insert examples with triggers paired with target labels
- Example: Image classifier trained to misclassify images containing specific pattern

How It Works:

- Attacker gains access to training data pipeline
- Injects malicious examples
- Model learns corrupted patterns
- Deployed model exhibits intended misbehavior

Risks:

- Compromised model reliability
- Hidden backdoors for exploitation
- Difficult to detect
- Persistent vulnerability

Mitigation:

- Data validation and sanitization
- Anomaly detection in training data
- Secure data pipelines
- Data provenance tracking
- Robust training techniques

Adversarial Evasion Attacks:

What It Is: Small, often imperceptible perturbations to inputs that cause models to make incorrect predictions.

How It Works:

- Craft inputs with subtle modifications
- Changes invisible or minor to humans
- Cause dramatic changes in model predictions
- Exploit model's learned decision boundaries

Examples:

- Adding noise to images to fool image classifiers

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- Modifying audio to bypass speech recognition
- Altering text to evade spam filters
- Physical adversarial examples (stickers on stop signs fooling autonomous vehicles)

Risks:

- Bypassing security systems
- Fraudulent access
- Safety hazards in autonomous systems
- Undermining model reliability

Types:

- **White-box attacks**: Attacker has full model access
- **Black-box attacks**: Attacker only observes inputs/outputs
- **Physical attacks**: Real-world adversarial examples

Mitigation:

- Adversarial training (training on adversarial examples)
- Input validation and sanitization
- Ensemble methods
- Detection mechanisms
- Robust model architectures

5.2 Traditional Security Concerns

Data Breaches:

- Unauthorized access to training data
- Exposure of sensitive information
- Regulatory and reputational consequences

Insider Threats:

- Malicious employees or contractors
- Accidental misuse
- Access control failures

Supply Chain Attacks:

- Compromised third-party components
- Malicious libraries or datasets

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- Vendor security failures

Infrastructure Vulnerabilities:

- Cloud platform security
- API vulnerabilities
- Network attacks

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6. RESPONSIBLE AI PRINCIPLES

6.1 Core Principles

Fairness:

Definition: AI systems should treat all individuals and groups equitably, without unjust discrimination.

Key Aspects:

- Equal treatment regardless of protected characteristics
- Avoiding discriminatory outcomes
- Addressing historical biases
- Ensuring equitable access to benefits

Implementation:

- Bias testing throughout lifecycle
- Fairness metrics and monitoring
- Representative training data
- Diverse teams and perspectives
- Regular fairness audits

Challenges:

- Multiple definitions of fairness (sometimes incompatible)
- Tradeoffs between fairness and accuracy
- Addressing intersectionality
- Context-specific determinations

Safety:

Definition: AI systems should operate reliably, not cause harm, and have appropriate safeguards.

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Key Aspects:

- Preventing physical harm
- Avoiding psychological or emotional harm
- Graceful failure modes
- Human oversight for high-risk decisions
- Emergency stop capabilities

Implementation:

- Comprehensive testing including edge cases
- Safety validation before deployment
- Monitoring for unsafe behavior
- Kill switches and override mechanisms
- Incident response plans

Considerations:

- Safety testing across diverse scenarios
- Real-world vs. test environment differences
- Unexpected use cases
- Cascading failures

Privacy:

Definition: AI systems should protect personal information and respect individuals' privacy rights.

Key Aspects:

- Data minimization
- Purpose limitation
- Transparency about data use
- User control and consent
- Security of personal data

Implementation:

- Privacy by design and default
- Privacy Impact Assessments
- Privacy-enhancing technologies

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- Clear privacy notices
- Data subject rights mechanisms

Framework: Based on Fair Information Practices (FIPs)

Transparency:

Definition: Making information available about AI use, operation, and decision-making.

Key Aspects:

- Disclosure of AI use
- Information about how systems work
- Documentation of capabilities and limitations
- Clear accountability structures
- Accessible information for stakeholders

Levels:

- **System-level**: What AI is used, for what purposes
- **Process-level**: How AI operates and makes decisions
- **Outcome-level**: Explanation of specific decisions

Implementation:

- Clear AI disclosure to users
- Model cards and system documentation
- Public reporting on AI use
- Stakeholder communication
- Transparency reports

Challenges:

- Balancing transparency with intellectual property
- Making technical information accessible
- Transparency to diverse audiences
- Proprietary vs. open systems

Explainability (Interpretability):

Definition: The ability to describe how an AI model works, explain its results and impacts, make the development and operation process accessible, and enable challenges to outputs.

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Key Aspects:

- Understanding how models make decisions
- Providing meaningful explanations for outputs
- Enabling stakeholders to understand AI behavior
- Facilitating debugging and improvement

Levels:

- **Global explainability**: How model works overall
- **Local explainability**: Why specific prediction was made
- **Counterfactual explainability**: What would need to change for different outcome

Techniques:

- Feature importance analysis
- LIME (Local Interpretable Model-agnostic Explanations)
- SHAP (SHapley Additive exPlanations)
- Attention visualization
- Decision trees for interpretability

Relationship to Transparency:

- **Transparency**: Making information available (what, when, where)
- **Explainability**: Making systems understandable (how, why)
- Complementary concepts that together build trust

Challenges:

- Accuracy-interpretability tradeoff
- Deep learning models inherently opaque
- Different stakeholders need different explanations
- No universal explanation method

Implementation:

- Choosing interpretable models when appropriate
- Post-hoc explanation methods for complex models
- Documentation of decision logic
- User-friendly explanation interfaces

Accountability:

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Definition: Clear assignment of responsibility for AI systems and their impacts, with mechanisms for oversight and redress.

Key Aspects:

- Defined roles and responsibilities
- Organizational accountability structures
- Oversight mechanisms
- Audit trails and documentation
- Redress and appeal processes

Components:

Organizational Responsibility:

- Board and executive oversight
- Clear governance structures
- Policies and procedures
- Resource allocation

Individual Accountability:

- Defined roles (developers, deployers, users)
- Training and competency requirements
- Performance metrics
- Ethical responsibilities

Mechanisms:

- Governance review boards
- Regular audits and assessments
- Incident investigation procedures
- Compliance monitoring
- External oversight

Implementation:

- Documented accountability frameworks
- Clear lines of authority
- Decision logs and audit trails
- Performance metrics and KPIs
- Regular reviews and updates

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Human-Centricity (Human Oversight):

Definition: AI should augment and empower humans, with appropriate human involvement in decisions, especially high-stakes ones.

Key Aspects:

- Humans remain in control
- AI serves human interests and values
- Meaningful human involvement in decisions
- Respect for human autonomy
- Human-in-the-loop for critical decisions

Levels of Human Oversight:

- **Human-in-the-loop**: Human reviews and approves each decision
- **Human-on-the-loop**: Human monitors and can intervene
- **Human-out-of-the-loop**: System operates autonomously (appropriate only for low-risk contexts)

Implementation:

- Override and intervention capabilities
- Clear escalation procedures
- Training for human operators
- Appropriate automation levels based on risk
- User control mechanisms

Considerations:

- Automation bias (over-reliance on AI)
- Deskilling of human operators
- Maintaining human expertise
- Balancing efficiency with oversight

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7. OPERATIONALIZING RESPONSIBLE AI

7.1 Strategies for Implementation

Embedding Principles into Practice:

Leadership Commitment:

- Executive sponsorship

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- Resource allocation
- Accountability at top levels
- Integration into strategy

Organizational Culture:

- Ethical awareness and training
- Encouraging responsible behavior
- Rewarding ethical practices
- Safe reporting of concerns

Policies and Procedures:

- Documented AI governance policies
- Standard operating procedures
- Review and approval processes
- Risk management frameworks

Technical Implementation:

- Bias testing and mitigation
- Privacy-enhancing technologies
- Security measures
- Monitoring and auditing systems

Stakeholder Engagement:

- Diverse perspectives in development
- User feedback mechanisms
- External audits and reviews
- Transparency with affected populations

7.2 Fairness Metrics and Testing

Approaches to Fairness Assessment:

Pre-deployment:

- Bias audits of training data
- Testing across demographic groups
- Fairness metric evaluation
- Red teaming and adversarial testing

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Post-deployment:

- Ongoing monitoring of outcomes
- Disaggregated performance metrics
- User feedback analysis
- Regular re-evaluation

Key Metrics:

- Error rate disparities
- Outcome distribution analysis
- Calibration across groups
- Fairness-aware performance measures

7.3 Building Trustworthy AI Culture

Elements:

Training and Education:

- AI ethics training for all employees
- Technical training on responsible AI
- Awareness of potential harms
- Best practices dissemination

Diverse Teams:

- Inclusive hiring practices
- Diverse perspectives in development
- Inclusive design processes
- Representation in leadership

Oversight Structures:

- AI ethics committees or boards
- Cross-functional review teams
- External advisory boards
- Independent audits

Continuous Improvement:

- Learning from incidents
- Regular policy reviews

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- Staying current with best practices
- Adapting to new risks and technologies

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8. CASE STUDIES OF AI HARMS

8.1 Amazon's Biased Hiring Tool

What Happened:

- Amazon developed AI to screen resumes and rank candidates
- Trained on 10 years of historical hiring data
- System learned to penalize resumes containing words like "women's" (e.g., "women's chess club")
- Downgraded graduates of all-women's colleges
- Discriminated against female candidates

Root Cause:

- Historical data reflected male-dominated tech industry
- Algorithm learned existing gender bias
- Training data not representative of desired outcomes

Outcome:

- Amazon scrapped the system
- Public embarrassment and reputational harm
- Highlighted risks of using biased historical data

Lessons:

- Historical data can perpetuate discrimination
- "Neutral" AI can learn and amplify bias
- Need for fairness testing before deployment
- Importance of diverse training data

8.2 Healthcare AI Disparities

What Happened:

- Various medical AI systems showing reduced accuracy for certain demographics
- Algorithms trained primarily on data from certain populations
- Performance degradation for underrepresented groups

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- Potential for misdiagnosis and improper treatment

Examples:

- Skin cancer detection AI less accurate on darker skin tones
- Pulse oximeters (aided by algorithms) overestimating oxygen levels in Black patients
- Diagnostic algorithms trained on non-diverse patient populations

Root Cause:

- Sampling bias in medical research and data
- Historical underrepresentation in clinical trials
- Lack of diversity in training datasets

Impact:

- Health disparities amplified
- Reduced quality of care for minorities
- Erosion of trust in medical AI

Lessons:

- Critical importance of representative medical data
- Need for testing across diverse populations
- High stakes of bias in healthcare
- Ethical obligation for equity in medical AI

8.3 Predictive Policing Bias

What Happened:

- AI systems used to predict crime hotspots and identify high-risk individuals
- Trained on historical crime data
- Perpetuated patterns of over-policing in minority communities
- Created feedback loops reinforcing bias

Issues:

- Historical data reflects biased policing practices
- More police presence in certain neighborhoods leads to more arrests
- More arrests generate more training data suggesting high crime
- Cycle continues and intensifies

Impact:

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- Disproportionate targeting of minority communities
- Civil rights concerns
- Reinforcement of systemic discrimination
- Erosion of community trust

Lessons:

- Historical bias in data leads to biased predictions
- Feedback loops can amplify initial biases
- Context matters—"objective" data may reflect subjective practices
- Need for fairness considerations in sensitive applications

8.4 Facial Recognition Errors

What Happened:

- Commercial facial recognition systems showing significantly higher error rates for people with darker skin tones, women, and younger/older individuals
- MIT study found error rates up to 34% higher for darker-skinned women
- Led to false arrests and wrongful accusations

Examples:

- Multiple cases of Black individuals falsely arrested due to misidentification
- Airport security and law enforcement deployment despite known inaccuracies

Root Cause:

- Training datasets predominantly featuring lighter-skinned faces
- Lack of diversity in data collection
- Insufficient testing across demographic groups

Impact:

- Civil rights violations
- Wrongful arrests and trauma
- Discriminatory outcomes in access control
- Public backlash and bans in some jurisdictions

Response:

- Some cities banned facial recognition for law enforcement
- Calls for moratoriums until accuracy improves

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- Regulatory scrutiny increased

Lessons:

- Accuracy disparities have serious real-world consequences
- Technology not always ready for deployment
- Need for rigorous fairness testing
- Importance of context-appropriate accuracy thresholds

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9. FRAMEWORKS FOR RESPONSIBLE AI

9.1 OECD AI Principles (Revisited with Module 2 Focus)

Applied to minimizing harms and promoting responsible AI:

1. **Inclusive Growth**: Ensuring AI benefits all populations, not just privileged groups
2. **Human-Centered Values**: Prioritizing fairness, dignity, and rights
3. **Transparency**: Enabling oversight and accountability for harmful impacts
4. **Robustness**: Preventing failures that cause harm
5. **Accountability**: Clear responsibility for addressing harms

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9.2 Fair Information Practices (FIPs) in AI Context

Notice: Inform individuals about AI use and data processing
Choice: Provide meaningful options regarding data and AI use
Access: Enable individuals to review and correct data
Integrity: Maintain accurate, secure data
Enforcement: Mechanisms to address violations

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10. KEY DEFINITIONS FOR MODULE 2

Bias: Systematic and unfair discrimination against certain individuals or groups.

Implicit Bias: Unconscious attitudes or stereotypes held by humans that influence AI development.

Sampling Bias: Training data that doesn't accurately represent the population the AI will serve.

Temporal Bias (Concept Drift): Model performance degradation when the world changes after training.

Overfitting: Model learns training data too well, including noise, failing to generalize.

Differential Privacy: Mathematical technique adding noise to protect individual privacy while preserving aggregate patterns.

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Homomorphic Encryption: Encryption allowing computations on encrypted data without decrypting.

De-identification: Removal or modification of identifying information (anonymization or pseudonymization).

Federated Learning: Training models across decentralized devices without exchanging raw data.

Model Inversion: Attack reconstructing training data by querying the model.

Model Extraction: Attack creating a copy of a model by observing inputs/outputs.

Data Poisoning: Manipulating training data to compromise model behavior.

Adversarial Evasion: Small input perturbations causing incorrect predictions.

Fairness: Equitable treatment without unjust discrimination.

Safety: Operating reliably without causing harm.

Privacy: Protecting personal information and respecting privacy rights.

Transparency: Making information available about AI use and operation.

Explainability: Ability to describe how a model works and explain its results.

Accountability: Clear assignment of responsibility with oversight and redress mechanisms.

Human-Centricity: AI augmenting humans with appropriate human involvement in decisions.

Individual Harms: Negative impacts on specific persons (discrimination, privacy violations, economic harm).

Group Harms: Impacts affecting demographic groups (systemic discrimination, representational harm).

Societal Harms: Broad impacts on society (misinformation, democratic threats, social division).

Environmental Harms: Negative impacts on the environment (energy consumption, carbon footprint).

Memorization Aids:

Four Types of Bias (I-S-T-O):

- **I**mplicit (human bias)
- **S**ampling (data bias)
- **T**emporal (drift)
- **O**verfitting (training issue)

Four PETs (D-H-D-F):

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- **D**ifferential privacy
- **H**omomorphic encryption
- **D**e-identification
- **F**ederated learning

Four AI Security Threats (I-E-P-E):

- Model **I**nversion
- Model **E**xtraction
- Data **P**oisoning
- Adversarial **E**vasion

Seven Responsible AI Principles (F-S-P-T-E-A-H):

- **F**airness
- **S**afety
- **P**rivacy
- **T**ransparency
- **E**xplainability
- **A**ccountability
- **H**uman-centricity

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